OBJECT DETECTION · SET PREDICTION · REAL TIME

NMS-Free Real-Time Detection

DETR → RT-DETR: moving duplicate suppression from post-processing into the loss, and the compute bottleneck into the architecture.

> **Central claim** — one-to-one Hungarian matching learns duplicate suppression, so NMS disappears. RT-DETR keeps that principle and re-engineers encoder, query learning, and decoder depth for real time.

42.0
DETR R50 AP · 41M params

53.1
RT-DETR R50 AP · 42M

108
FPS · T4 · TensorRT FP16

34.8
AP_S · small-object gap

Problem — Anchors · duplicate boxes · NMS cost
› **DETR** — Set prediction · optimal one-to-one matching
› **RT-DETR** — Efficient encoder · better queries · tunable depth

1 · CITE
every number is either a **cited paper table** or labelled as our own measurement — the two are never mixed.

2 · REPRODUCE
the encoder-variant latency **order** (D_S5 cheapest) reproduces paper Table 3 exactly on our hardware.

3 · MEASURE
our NMS cost model is benchmarked **end-to-end with the plugin included**, matching the paper's conf / IoU trade-off direction.

4 · VERIFY
the asymmetry the papers teach: matching cost in **probabilities**, loss in **log-likelihoods** — L1 penalises a large box 9.44× more, GIoU 1.000×.

5 · SCOPE
our re-drawings are **synthetic and CPU-scale**; real-time claims stay with the paper's T4 / TensorRT FP16 benchmark.

Carion et al., ECCV 2020 — arXiv:2005.12872 · Lv et al., CVPR 2024 — arXiv:2304.08069
Code: github.com/facebookresearch/detr · github.com/lyuwenyu/rt-detr
Figures: our Day 07–08 measurements (synthetic, CPU) unless a paper table is cited.

01 · STARTING POINT

What changed in DETR?

Conventional detectors emit many proposals or anchors and let **NMS heuristics** remove duplicates. DETR predicts the object **set** in one pass.

Backbone features › Transformer global context › 100 queries parallel decode

Matching before the loss

The Hungarian algorithm finds the **globally optimal one-to-one** assignment between prediction slots and ground-truth objects, so duplicates become structurally impossible.

$$L = L_{class} + \lambda_1 L_1 + \lambda_G L_{GIoU}$$

Matching cost uses **probabilities** (bounded, commensurable with the box term); the loss uses **log-likelihoods**. Under the same 15% relative error, L1 penalises a large box **9.44×** more than a small one, while GIoU stays at **1.000×**.

02 · THE NMS DILEMMA

Two thresholds, one trade-off

Paper Table 1 (YOLOv8, T4 · TensorRT FP16): raising the confidence threshold removes candidates — **faster but less accurate**; raising the IoU threshold suppresses fewer boxes — **more accurate but slower**.

CONTROL	AP	NMS
conf 0.001 → 0.05	52.9 → 51.2	2.36 → 1.06 ms
IoU 0.5 → 0.8	52.1 → 52.8	2.24 → 2.46 ms

YOLO papers report model speed **excluding** NMS time, so RT-DETR builds an end-to-end benchmark (§3.2) with the NMS plugin included.

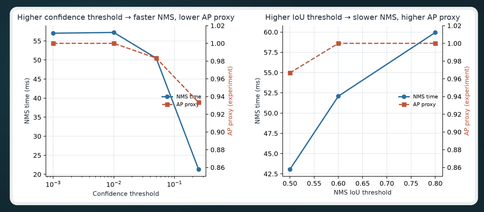


Fig. 1. NMS cost model — our measurement; direction matches paper Table 1.

03 · BOTTLENECK → DESIGN

Why is AIFI only on S5?

In Deformable-DETR the encoder consumes **49% of GFLOPs** for **11% of AP** (a second-hand statistic, not our measurement). RT-DETR re-architects it:

AIFI self-attention on **S5 only** (400 tokens)
CCFF conv-based cross-scale fusion (linear cost)

Attention cost grows with **M²**: 8400 candidate tokens vs 400 on S5 is a **441×** gap. Variant **D_S5**: latency **–35%** and AP **+0.4**.

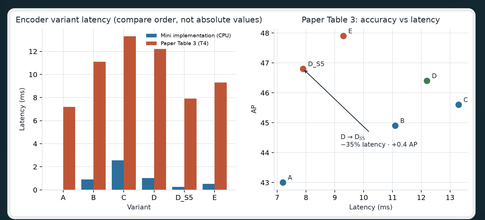


Fig. 2. Encoder variants — paper Table 3; our latency order reproduces it exactly.

04 · QUERY QUALITY

What does uncertainty change?

U = ‖P − C‖ — the mismatch between the **location** and **classification** prediction of a feature — enters the loss, so the encoder learns features that agree in both.

AP 47.9 → **48.7** · **Prop cls** 0.35% → **0.82%** · **Prop both** 0.30% → **0.67%**

Selection ≠ objective: the public top-K still ranks by class score (`rtdetr_decoder.py:491`) — uncertainty changes the **loss**, not the ranking.

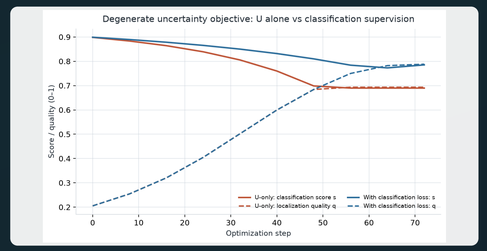


Fig. 3. Uncertainty alone has a degenerate optimum — our measurement.

05 · INFERENCE CONTROL

Tune speed with depth

Auxiliary heads train **every** decoder layer, so inference can stop early **without retraining**. Paper Table 5 (Det 6): 6 layers 53.1 AP / 9.3 ms → 5 layers 53.0 / 8.8 ms → 1 layer 49.1 / 7.0 ms.

Cost per layer is nearly constant (0.4–0.5 ms) while the accuracy gain shrinks with depth (+2.2 AP at layer 1→2, only +0.1 at 5→6).

eval_idx = which layer · **max_det = output cap** · **lower max_det ≠ faster**

06 · RESULTS

Accuracy and speed together

MODEL	AP	AP_S	AP_L	FPS
DETR R50	42.0	20.5	**61.1**	28
RT-DETR R50	**53.1**	**34.8**	**70.0**	**108**

Same parameter count as Faster R-CNN-FPN+, but the scale profile flips: **AP_S –5.5**, **AP_L +7.8**. RT-DETR: **+2.2 AP and 21× faster** than DINO-Deformable (5 → 108 FPS).

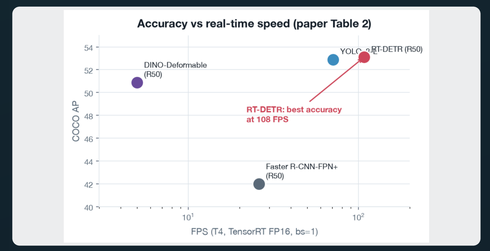


Fig. 4. Accuracy vs real-time speed — paper Table 2 (NMS included for YOLO baselines).

07 · OPEN LIMITS

What remains unresolved

- **Small objects stay weak** — AP_S 34.8 (–0.5 vs YOLOv8-L; R101 –0.9): the cost of restricting global attention to S5 while keeping multi-scale features.
- **49% / 11% is a second-hand statistic** — cited, not measured by us.
- **138% / 120% are relative counts**, not percentage-point gains.
- **2D COCO only** — depth, BEV and multi-view remain untested.

08 · STUDY PROCESS

Claim → evidence → boundary

8 study days, 16 deep-dive documents, 4 runnable demos, 22 charts, **0 citation errors**. Each claim was checked against the original table or equation, then probed with our own matching / NMS / attention / query-loss experiments.

Those checks are **CPU, synthetic, direction-only** — they are not reproductions of the paper's COCO/GPU numbers.

Next

GLIP + SAM · **CLIP** · **LLaVA ×2** · **VL-JEPA** · **3D: DETR3D → PETR → Sparse4D**

Move duplicates into the loss · compute into the architecture · initialization into quality

09 · EVIDENCE AT A GLANCE

Our own measurements

Four checks that support the design decisions, all reproduced locally:

- **A** greedy vs optimal matching: duplicates **31.6%**, greedy never wins
- **B** attention cost: 8× tokens → **42.3×** time (theory 64×)
- **C** uncertainty alone: s 0.90 → **0.690** (degenerate)
- **D** encoder latency order = paper order

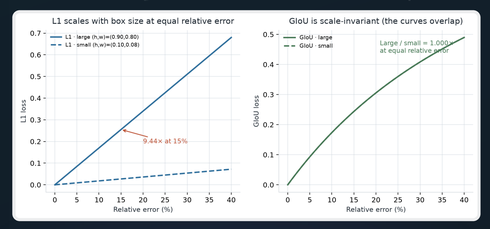


Fig. 5. L1 scales with box size, GIoU does not — our measurement.